ECON 381: Midterm 1

Professor Muyang Ren

Fall 2026

Name: _____

Instructions

1. Write your name on this exam, your note sheet, and every answer sheet.
2. **Do not open this exam until instructed to do so.**
3. Silence your cell phone and place it in your bag, not in your pocket.
4. This exam is worth 100 points, with 5 additional bonus points available. The maximum recorded score is 100.
5. Check that you have all 7 pages. You have 75 minutes to complete the exam.
6. You may use one 8.5×11 sheet of handwritten notes, on both sides, prepared before the exam. Turn in your note sheet with your exam and answer sheets.
7. Write all answers and supporting work on separate blank sheets of paper. Do not write answers on the exam pages. Clearly number each answer page, label each answer with its section, question number, and part. For each multiple-choice question, write the question number and the corresponding one answer letter.
8. Use other blank sheets for scratch work. Clearly cross out any work you do not want graded in your answer sheets.
9. One standalone, non-programmable calculator is allowed. All other electronic devices are prohibited. For confidence intervals, round the standard normal critical values discussed in class to two decimal places.
10. Sign the honor pledge below. At the end of the exam, turn in this exam, your answer sheets, and your note sheet.
11. **Good luck!**

I pledge on my honor as a University of Tennessee student that I will neither give nor receive aid on this examination.

Signature: _____

1. Multiple Choice Questions (30 points total; 10 questions; 3 points each)

Instructions: On your separate answer sheets, write each question number and **one** answer letter (a, b, c, or d). No explanation is required. Selecting more than one answer earns no credit for that question.

1. For an i.i.d. sample with a finite mean, which statement expresses the Law of Large Numbers?

   (a) $\bar{Y}$ equals μ_Y for every sample size.
   (b) The probability that $|\bar{Y} - \mu_Y|$ exceeds any fixed positive number approaches zero as n grows.
   (c) The population mean changes as the sample size grows.
   (d) Every individual observation approaches μ_Y as n grows.

2. The fitted line is $\widehat{\text{TestScore}} = 698.9 - 2.28\,STR$. If the student–teacher ratio decreases by 2, the fitted test score

   (a) increases by 4.56 points.
   (b) decreases by 2.28 points.
   (c) decreases by 4.56 points.
   (d) increases by 2.28 points.

3. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ_Y and finite, positive variance σ_Y^2. The Central Limit Theorem states that, as n grows,

   (a) each Y_i becomes normally distributed.
   (b) $\bar{Y}$ equals μ_Y with certainty.
   (c) the population distribution must become symmetric.
   (d) $\sqrt{n}(\bar{Y} - \mu_Y)/\sigma_Y$ approaches a standard normal distribution.

4. In $\text{TestScore}_i = \beta_0 + \beta_1 STR_i + u_i$, the population regression error u_i represents

   (a) the fitted test score from the estimated sample line.
   (b) the estimated slope of the sample regression line.
   (c) other influences on test scores not represented by STR_i in the population line.
   (d) the student–teacher ratio in district i.

5. For a large random sample, which is the approximate two-sided 99% confidence interval for the population mean?

   (a) $\bar{Y} \pm 1.64\,SE(\bar{Y})$.
   (b) $\bar{Y} \pm 1.96\,SE(\bar{Y})$.
   (c) $\bar{Y} \pm 2.58\,SE(\bar{Y})$.
   (d) $\bar{Y} \pm 2.58\,s_Y$.

6. What does $E[\text{TestScore} \mid C = \text{Los Angeles}]$ mean, where C denotes county?

   (a) The population average test score among districts in Los Angeles County.
   (b) The population average test score across all counties combined.
   (c) The probability that a district is in Los Angeles County.
   (d) The sample average test score among districts in Los Angeles County.

7. For i.i.d. sampling, holding population dispersion and the confidence level fixed, increasing the sample size generally makes the standard error of the sample mean and the confidence interval width

   (a) both larger.
   (b) smaller and larger, respectively.
   (c) unchanged.
   (d) both smaller.

8. Ordinary least squares chooses the intercept and slope to minimize

   (a) $\sum_{i=1}^{n}(X_i - \bar{X})^2$.
   (b) $\sum_{i=1}^{n}[Y_i - (b_0 + b_1 X_i)]^2$.
   (c) $\sum_{i=1}^{n}(Y_i - \bar{Y})^2$.
   (d) $\sum_{i=1}^{n}[Y_i - (b_0 + b_1 X_i)]$.

9. A very large positive covariance between X and Y does not necessarily imply that their correlation is close to one because

   (a) correlation must be negative whenever covariance is positive.
   (b) covariance is always between -1 and 1.
   (c) covariance depends on measurement units, while correlation divides covariance by both standard deviations.
   (d) correlation equals covariance for every pair of variables.

10. The standard error of the regression (SER) describes

    (a) the typical residual size, measured in the units of Y.
    (b) the standard error of the sample mean $\bar{Y}$.
    (c) the estimated slope of the fitted line.
    (d) the fraction of sample variation in Y captured by the fitted line.

2. True or False Questions (20 points total; 4 questions; 5 points each)

Instructions: On your separate answer sheets, label each part and write **T** or **F**. You must justify each answer. Full credit requires a correct answer and justification.

(a) A 95%-level confidence interval means that, across repeated samples, the population mean takes different values and falls into a fixed interval with probability 95%.

(b) The sample standard deviation s_Y describes the spread of individual outcomes, while $SE(\bar{Y})$ estimates the spread of sample averages across repeated random samples of the same size.

(c) If the population correlation between X and Y equals 1, then observations drawn from this population lie on a straight line with a positive slope.

(d) If Y_i is multiplied by 2 while X_i is unchanged, then SER and R^2 will be multiplied by 2.

3. Analytical Question (50 points total)

Instructions: Write your answers and show all work on your separate answer sheets. Label each part (a)–(i). Use the same three observations throughout. Parts (a)–(e) concern the population mean of Y; parts (f)–(i) concern the regression of Y on X with an intercept.

Consider this hypothetical random sample of three observations:

Observation i	X_i	Y_i
1	2	16
2	3	10
3	4	10

(a) (4 points; 2 each) Compute the sample averages $\bar{X}$ and $\bar{Y}$.

(b) (5 points) Compute the sample standard deviation s_Y, using the sample variance with divisor $n-1$. You may leave a square root in your answer.

Hint: $s_Y = \sqrt{\dfrac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2}$.

(c) (4 points) Compute the standard error of the sample average, $SE(\bar{Y})$.

(d) (8 points; 4 each) Compute the 90% and 95% confidence intervals for the population mean μ_Y. Round the critical values to two decimal places and report endpoints to two decimal places.

(e) (4 points) With only these three observations, does the procedure used to construct the 95% confidence interval in part (d) necessarily cover the population mean in 95% of repeated samples? Explain.

Analytical Question (continued)

The data remain $(X_i, Y_i) = (2, 16), (3, 10), (4, 10)$. Estimate

$$Y_i = \beta_0 + \beta_1 X_i + u_i.$$

(f) (10 points) Compute the OLS slope estimator $\hat{\beta}_1$. Show the numerator and denominator of the slope formula using deviations from the sample means.

(g) (6 points) Compute the OLS intercept estimator $\hat{\beta}_0$ and write the fitted regression equation.

(h) (6 points; 1 per entry) Compute the three fitted values $\hat{Y}_i$ and three residuals $\hat{u}_i$. Copy the table onto your separate answer sheets and fill in the fitted values and residuals there.

i	X_i	Y_i	$\hat{Y}_i$	$\hat{u}_i$
1	2	16		
2	3	10		
3	4	10		

(i) (3 points) Compute R^2. Show the total sum of squares (TSS) and the residual sum of squares (RSS) used in your calculation.

4. Bonus Question (5 points)

Instructions: Write your answers on your separate answer sheets and label parts (a) and (b). Explain your reasoning using the OLS formulas or minimization problem discussed in class.

A researcher observes the same outcome for all three observations, while the regressor varies:

Observation i	X_i	Y_i
1	3	12
2	2	12
3	4	12

The researcher fits $Y_i = \beta_0 + \beta_1 X_i + u_i$ with both an intercept and a slope.

(a) (3 points) Is the OLS slope well-defined for this sample? Explain using the numerator and denominator of the slope formula. If it is well-defined, compute the slope. Also, would your answer change if instead $X_1 = X_2 = X_3 = 3$ and the outcomes were $Y_1 = 16$, $Y_2 = 10$, and $Y_3 = 10$? Explain.

(b) (2 points) Is the usual R^2 well-defined for this sample? Explain using the numerator and denominator of the R^2 formula. If it is well-defined, compute the R^2.